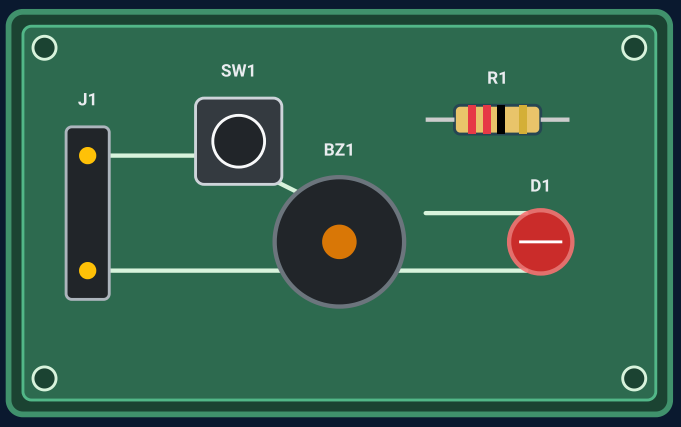
KICAD PCB DESIGN PROJECT

# EMERGENCY PANIC ALARM

Schematic • PCB Layout • 3D View • Gerber Output

**Saksham Tyagi**
ECE | VLSI Specialist

Hardware Design | KiCad v8.0



## PROJECT OVERVIEW

The **Emergency Panic Alarm** is a robust 5V alert circuit engineered for immediate visual and acoustic warning signaling. Designed for reliability and swift activation, pressing the momentary switch instantly triggers the high-decibel piezo buzzer and high-luminosity warning LED.

INPUT STAGE
**5V INPUT**
DC Power connector via J1 header interface

TRIGGER STAGE
**PUSH BUTTON**
Momentary normally-open tactile switch SW1

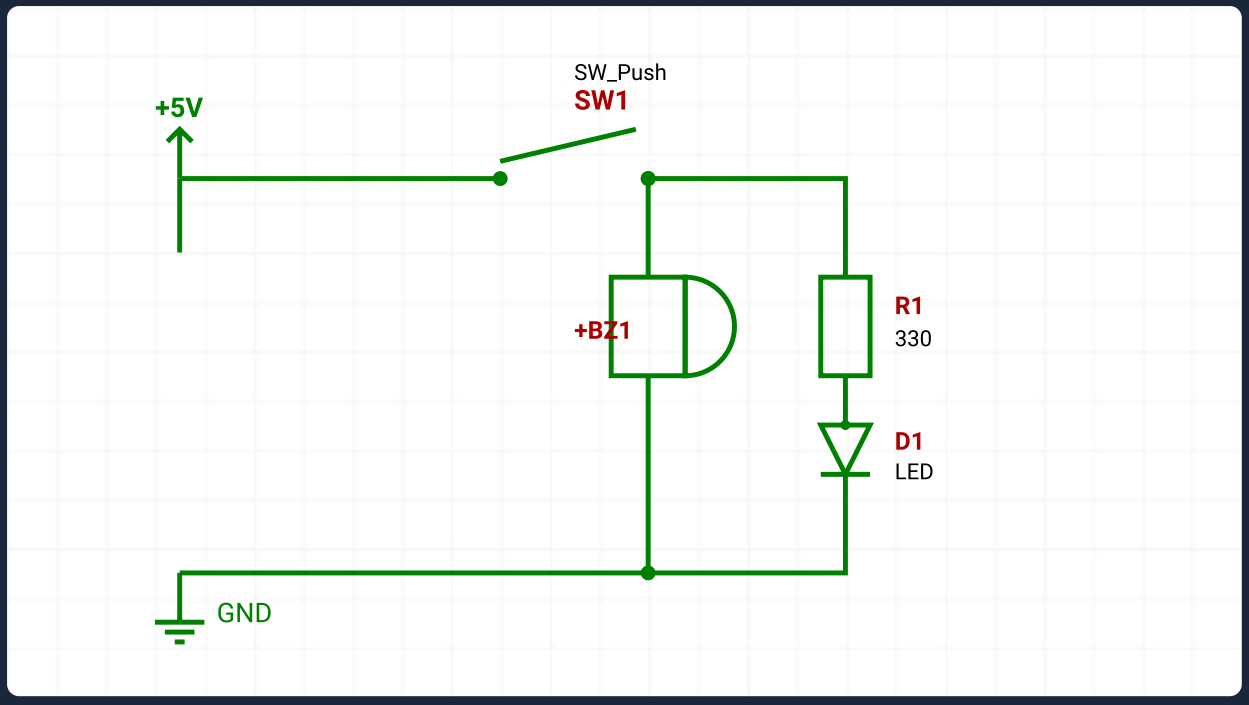
OUTPUT STAGE
**ALERT ACTIVE**
Simultaneous Buzzer audio & LED indicator

## BILL OF MATERIALS (BOM)

| Reference | Description / Specification | Qty | Footprint Type |
|---|---|---|---|
| **J1** | 2-pin 5V DC Power Input Connector | 1 | PinHeader_1x02_P2.54mm |
| **SW1** | Normally-Open Push Button Switch | 1 | SW_PUSH_6mm |
| **BZ1** | 5V Active Piezo Acoustic Buzzer | 1 | Buzzer_12x9.5mm_P7.6mm |
| **D1** | 5mm High-Brightness Red LED | 1 | LED_D5.0mm |
| **R1** | 330 Ω Current Limiting Resistor (1/4W) | 1 | R_Axial_DIN0207_L6.3mm_D2.5mm |

# CIRCUIT SCHEMATIC

Complete circuit schematic illustrating power distribution flags, switch activation logic, and parallel alert component driving.

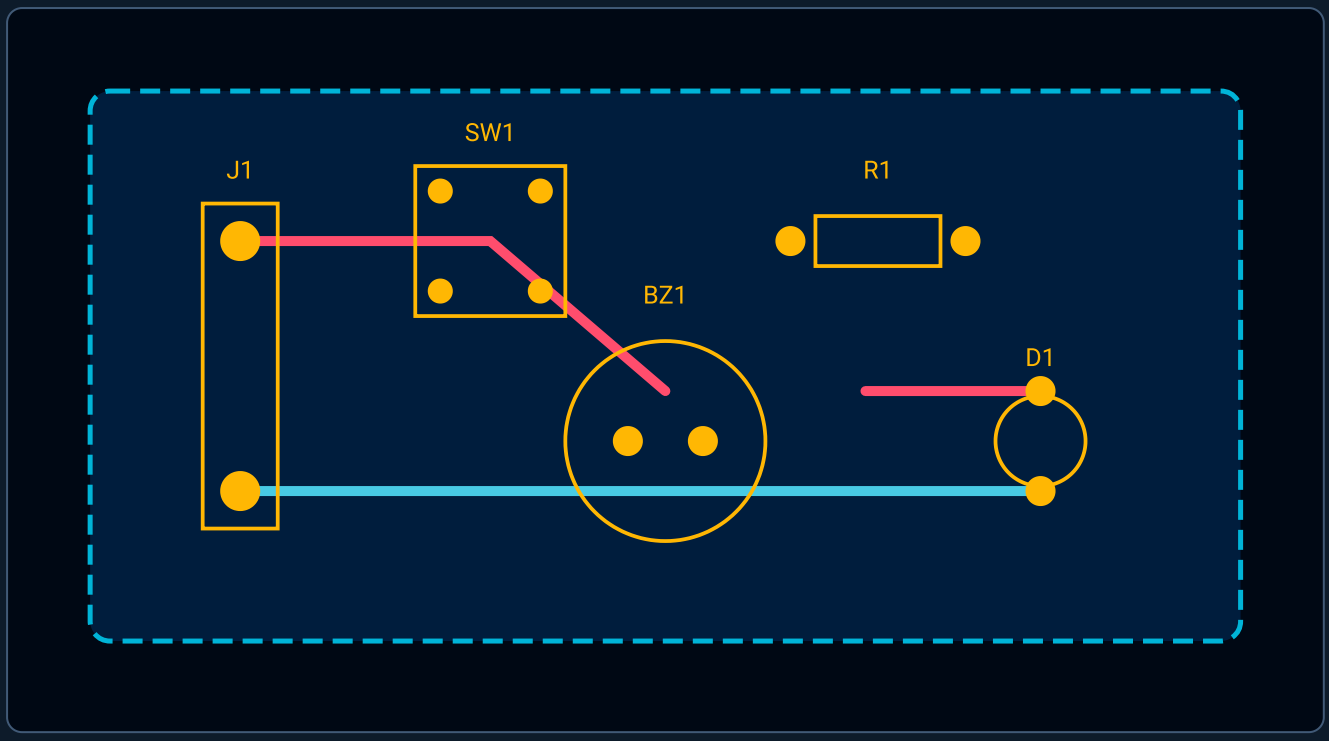



PWR_FLAG Verification: Passed

ERC Status: 0 Errors / 0 Warnings

# PCB LAYOUT & ROUTING

2-layer printed circuit board design with optimized copper traces, robust ground plane pours, and zero unrouted connections.




**Top Layer (F.Cu)**
Signal routing & components

**Bottom Layer (B.Cu)**
Solid GND plane fill

**DRC Status**
0 Violations / 0 Unrouted

# MANUFACTURING OUTPUT (GERBER FILES)

Production-ready Gerber and NC drill files exported for standard RS-274X fabrication.

```
📁 Gerber_Output_Files/
├── 📄 Emergency_Panic_Alarm-F_Cu.gbr        (Top Copper Layer)
├── 📄 Emergency_Panic_Alarm-B_Cu.gbr        (Bottom Ground Plane)
├── 📄 Emergency_Panic_Alarm-F_Mask.gbr      (Top Solder Mask)
├── 📄 Emergency_Panic_Alarm-B_Mask.gbr      (Bottom Solder Mask)
├── 📄 Emergency_Panic_Alarm-F_Silks.gbr     (Top Silkscreen)
├── 📄 Emergency_Panic_Alarm-Edge_Cuts.gbr   (Board Outline)
├── 📄 Emergency_Panic_Alarm-PTH.drl         (Plated Drill Holes)
└── 📄 Emergency_Panic_Alarm-NPTH.drl        (Non-Plated Mounting)
```

| Parameter | Specification |
|---|---|
| Board Finish / Material | FR-4 Standard / HASL Lead-Free |
| Board Thickness | 1.6 mm |
| Copper Weight | 1 oz (35 µm) Outer Layers |

VERIFIED & COMPLETE

# PROJECT COMPLETE

Emergency Panic Alarm PCB Layout & Schematic Verification Successfully Executed.

DESIGNED & DEVELOPED BY

**Saksham Tyagi**

ECE | VLSI

Designed in KiCad

Ready for Fabrication